MASTER OF ARTS IN EDUCATION - THESIS MANUSCRIPT

APPROVED RESEARCH TITLE

Data-Driven Instructional Supervision and Digital Readiness of Public Secondary School Heads in the Division of Man

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ABSTRACT

This study investigates the extent of data-driven instructional supervision practices and the level of digital readiness among public secondary school heads in the Division of Manila during Academic Year 2025-2026. Employing a descriptive-correlational research design, the study surveyed 32 public secondary school principals, 128 department heads, and 340 master teachers across six congressional districts in Manila.

Findings indicate that school heads demonstrate a high level of commitment to classroom observation analytics (M = 4.32, SD = 0.48) and learning outcome monitoring (M = 4.26, SD = 0.51), while digital infrastructure integration and predictive learner intervention scored moderately high (M = 3.84, SD = 0.62). Pearson product-moment correlation analysis revealed a significant positive relationship ($r = 0.76$, $p < 0.001$) between digital readiness and supervisory effectiveness.

CHAPTER 1: THE PROBLEM AND ITS BACKGROUND

1.1 Introduction and Rationale

Contemporary educational leadership increasingly relies on empirical school data and digital supervisory platforms to elevate teaching quality in urban public school systems. Within the Department of Education (DepEd) National Capital Region, secondary school heads are mandated to lead instructional supervision under the Philippine Professional Standards for School Heads (PPSSH).

However, bridging administrative data collection and actionable pedagogical coaching remains a persistent challenge. This manuscript examines how public secondary school heads synthesize classroom observation records, assessment analytics, and digital coaching tools to support faculty instructional competence.

CHAPTER 1 & CHAPTER 2: STATEMENT OF THE PROBLEM & FRAMEWORK

1.2 Statement of the Problem

This study aimed to determine the relationship between data-driven instructional supervision and digital readiness of public secondary school heads in the Division of Manila during AY 2025-2026. Specifically, it sought answers to the following research questions:

1. What is the demographic and professional profile of the respondent school heads in terms of administrative experience, highest educational attainment, and supervisory training hours?

2. To what extent do public secondary school heads practice data-driven instructional supervision across: (a) Classroom Observation & Instructional Diagnostics, (b) Curriculum Alignment & Assessment Analytics, (c) Collaborative Coaching & Feedback, and (d) Professional Learning Community (PLC) Facilitation?

3. What is the level of digital readiness of the respondents in terms of technological competence, digital governance, and data literacy?

4. Is there a significant relationship between the extent of data-driven instructional supervision and the level of digital readiness of public secondary school heads?

5. Based on the findings of the study, what Digital Instructional Supervision Action Plan may be proposed?

1.3 Theoretical and Conceptual Framework

The study is anchored on Mandinach and Gummer (2016) Data Literacy for Educators Framework and the DepEd Philippine Professional Standards for School Heads (DepEd Order No. 24, s. 2020). Using the Input-Process-Output (IPO) model, school head profiles, supervisory domains, and digital readiness indicators serve as inputs analyzed through descriptive and inferential statistics to formulate an institutional Digital Supervision Enhancement Program.

CHAPTER 3: RESEARCH METHODOLOGY & STATISTICAL TREATMENT

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

This research utilized a quantitative descriptive-correlational design to systematically describe the current supervisory practices and digital readiness of public secondary school heads and to test the strength and direction of association between the two constructs without experimental manipulation.

3.2 Population, Sample Size, and Sampling Technique

Using stratified proportional random sampling across the six congressional districts of the Division of Manila, a total of 32 public secondary school principals and 128 department heads participated as primary respondents, complemented by 340 master teachers who validated supervisory practices.

3.3 Research Instrument and Validation

A researcher-developed survey questionnaire structured on a 5-point Likert scale was content-validated by five (5) Graduate School experts in Educational Administration and Supervision. Pilot testing yielded a Cronbach alpha reliability coefficient of 0.93 for the Data-Driven Supervision scale and 0.91 for the Digital Readiness scale.

3.4 Statistical Treatment of Data

Summary of Statistical Tools Applied

1. Frequency and Percentage Distribution - To describe the professional profile of respondents.
2. Weighted Mean and Standard Deviation - To quantify the extent of data-driven supervision and digital readiness.
3. Pearson Product-Moment Correlation (r) - To test the significant relationship between supervisory practices and digital readiness at the 0.05 alpha level.
4. One-Way Analysis of Variance (ANOVA) - To test significant differences across respondent groups.